

Phase 1 Documentation

This is the phase where it started everything, I took some experience, architecture, and some code from [D4C-Port-Scanner](#) which was a previous project developed. I wanted to take that base, going beyond and develop this Automated Penetration Tool.

The Phase 1 is the first step of the hacking, Reconnaissance, gathering the information needed for perform a pen-test. Reconnaissance module handles boundary validation, port scanning, and service identification through a strict, decoupled Python architecture.

A. Phase 1 Settings and File Structure:

- **init.py:** Initializes **phase1_recon** as a standard Python package.
- **main.py:** Functions as the primary command-line execution script to launch scans.
- **models.py:** Uses Python Data Classes to enforce a strict schema for ports, protocols, and banners. It specifically filters for “**is_open**” or **state == "open"** to prevent exporting closed ports into the final JSON output.
- **scope_guard.py:** Enforces strict boundaries by validating target IPs (such as 127.0.0.1/32) against authorized CIDR ranges before scanning begins.
- **scanner.py:** The core port discovery engine. It transitioned from a pure `asyncio.Semaphore` setup to utilizing the `python-nmap` library wrapped in `asyncio.to_thread()` to maintain non-blocking asynchronous execution.

B. Problems Encountered & Fixes:

Problem Faced:	Implemented Fix:
Bloated JSON Output: The scanner initially exported data for all ports, including closed ones, creating bloated hand-off files.	Refined Data Models: Updated the <code>models.py</code> logic to drop irrelevant data and only export ports matching an open state.
Nmap Blocking Async Loop: Dropping a synchronous Nmap scan into the async orchestrator would block the entire Python event loop	Background Threading: Wrapped the Nmap execution inside <code>asyncio.to_thread()</code> to preserve the high-concurrency architecture
Nmap Executable Not Found: Running the scanner threw a <code>PortScannerError</code> because only the Python wrapper was installed	System PATH Update: Installed the official Windows Nmap binary, added its directory to the system PATH environment variable, and restarted the terminal

C. Testing Results:

On the terminal, run the following command on the terminal:

```
py -3 -m phase1_recon.main --target 127.0.0.1
```

1.1. Initial output:

```
PS C:\Users\ZackU\Desktop\Auto-Pentest> py -3 -m phase1_recon.main --target 127.0.0.1
[*] Target 127.0.0.1 is in scope. Starting scan (nmap default ports)...
█
```

1.2. Final output:

```
[+] Found 7 open port(s) on 127.0.0.1:
  135/TCP - msrpc (Microsoft Windows RPC)
  445/TCP - microsoft-ds (unknown version)
  902/TCP - vmware-auth (VMware Authentication Daemon 1.10)
  912/TCP - vmware-auth (VMware Authentication Daemon 1.0)
  5000/TCP - http (Werkzeug httpd 2.0.1)
  8080/TCP - http (Apache httpd 2.4.49)
  9080/TCP - glrpc (unknown version)
[+] Results saved to C:\Users\ZackU\Desktop\Auto-Pentest\data\scanner_results\127_0_0_1.json
█
```

1.3. Logs on Docker:

2026-08-30 21:01:47	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:01:47 +0000]	"GET / HTTP/1.0" 200 45
2026-08-30 21:01:47	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:01:47]	"GET / HTTP/1.0" 200 -
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"GET / HTTP/1.0" 200 -
2026-08-30 21:03:44	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:03:44 +0000]	"GET / HTTP/1.0" 200 45
2026-08-30 21:03:44	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:03:44 +0000]	"POST /sdk HTTP/1.1" 404 196
2026-08-30 21:03:44	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:03:44 +0000]	"GET /nmaplowercheck1788141824 HTTP/1.1" 404 196
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"GET /nmaplowercheck1788141824 HTTP/1.1" 404 -
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"POST /sdk HTTP/1.1" 404 -
2026-08-30 21:03:44	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:03:44 +0000]	"GET /evox/about HTTP/1.1" 404 196
2026-08-30 21:03:44	test_target/legacy-httpd	172.18.0.1	-	[31/Aug/2026:02:03:44 +0000]	"GET /HNAP1 HTTP/1.1" 404 196
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"GET /HNAP1 HTTP/1.1" 404 -
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"GET /evox/about HTTP/1.1" 404 -
2026-08-30 21:03:44	test_target/mock-target	172.18.0.1	-	[31/Aug/2026 02:03:44]	"GET / HTTP/1.0" 200 -

1.4. Result stored:

After running the scanner, the results of the findings are stored in a file using the format of the target IP address in **json** format for use in future phases.

```
{ } 127_0_0_1.json X
data > scanner_results > { } 127_0_0_1.json > ...
1  {
2    "target": "127.0.0.1",
3    "ip_address": "127.0.0.1",
4    "scan_timestamp": "2026-08-31T02:01:40.848313",
5    "open_ports": [
6      {
7        "port": 135,
8        "protocol": "TCP",
9        "state": "open",
10       "service": {
11         "name": "msrpc",
12         "banner": "Microsoft Windows RPC",
13         "version": "Microsoft Windows RPC"
```